

Homework Classical Physics, Fall 2023

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Problem Set 01.4 (L01 & L02)

Due on Thursday, 14. May 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - velocities (4 points)

- a)** You ride a bike a certain distance uphill at 10 km/h and then the same distance downhill at 20 km/h - what is your average speed?
- b)** You then ride your bike a certain time at 10 km/h and then the same time for 20 km/h - what is your average speed?
- c)** A plane flies through the air with speed v . After the distance d it turns around and flies back to its starting position. A wind is blowing with speed w in the direction of the plane for the first part and against the direction of the plane when returning. What is the total time the plane needs to go forth and back? Do the gain and loss due to the wind compensate?